The courtyard as the family's haven · a study in phases · Phase 0: the facts · Taif / Palm House V10.0

The owner's brief (10 Nov 2026): breeze, water, tree, shade, winter sun, views, privacy, calm, beauty, easy upkeep, all working together; find every mistake. **Locked:** the site and the outline, the structure, the stair and the lift, the settled rooms, the malqaf L5. **Open:** the seating, the water and the tree, the outlet (A or B), the sink window, the courtyard's doors and windows, the arch infill, the lighting, the floor, the planting, the drainage. **Owner's preferences:** no TV; every seat with a back, 5–6 seats; open to the sky; level floor, movable furniture; render and paint, porcelain floors; simple switches; privacy only from the guest route and the majlis side. **Amber = an assumption.** Phase 0 cites the files; Phase 1 checks them.

The courtyard as designed (V10.0 + the owner's review of 10 Nov 2026)

Item	Fact	Where
Courtyard on axes	an L: x 4.25–8.95 × y 6.5–13.0 less the malqaf core (x 4.4–5.8, y 6.5–8.667): **27.19 m²**	data/rooms_gf.csv 2
Courtyard clear (finished faces)	4.50 × 6.30 = 28.35 m² less the core's footprint (1.6 × 2.17 = 3.47): ≈ **24.88 m²**; the colonnade under the projection 1.575 wide (x 4.35–5.925), the open sky 3.15 × 6.5 on axes	courtyard_layout.typ 30–38 · sections.typ pj (depth 1.55, sky 3.15)
Walls round it	north: the women's majlis glass (+2.9 head) under the family living's W2; south: the living glass to +3.32 under W3; east: the kitchen wall two storeys to the roof (+7.3) and the 1.10 guard; west: the projection's soffit +3.27 on the colonnade, the gallery's W1 above (crown +6.60), the roof +7.3	params.yaml 11, 130 · structure.typ (soffit +3.27)
Malqaf chamber	the core x 4.4–5.8 × y 6.5–8.667 on axes (faces x 4.3–5.95, y 6.4–8.767; the east wall 0.30 on the column line since 10 Nov 2026); inside: the dust trap, the filter cabinet + EC fan + silencer, the cooler's space; the door 1.0 × 2.4 south onto the covered strip	air_system.typ 25, 31–34 · design_gf.typ 80, 138
Outlet A (as built)	1.4 × 1.0 on the east face, y 7.1–8.5, +0.1…+1.1, adjustable timber louvres aimed south-east (the air study: "the louvres aim the jet along the seating"); free area 0.84 m²; ≈ 1.79 m³/s and 2.1 m/s at the louvres at the summer-evening mean wind, 0.85 m³/s in a gentle wind, 0.49 m³/s in calm (stack)	air_system.typ 30, 50–51, 67, 228
Outlet B (studied, not adopted)	the same 1.4 × 1.0, raised to +0.60…+1.60, centred in the bay y 6.9–8.3, the louvres fixed 30° south-east with splayed reveals: over the	malqaf_outlet_B.typ 36
Seating (10 Nov 2026)	the level zone x 5.925–8.85 × y 7.1–12.3 (12.09 m² free of the enclosure): sofa A 1.5 × 0.85 facing W, sofa B 1.4 × 0.85 facing N, two armchairs 0.75 at the colonnade edge facing E, a 0.7 × 0.5 table; 6 seats, every seat with a back; no TV	courtyard_layout.typ 46–52 · design_gf.typ 79
Colonnade and arches	the edge beam on x 5.8 (C21 at y 6.5, C22 at 13.0; the first arcade column at 8.667 absorbed in the core, the second at 10.833, Ø0.25 rendered); three semicircular arch plates under the soffit +3.27, springing ≥ +2.31; the bays 2.167 on axes; the first bay filled by the core (the outlet in it)	character.yaml 79–81 · structure.typ X18 · design_gf.typ 74–80
Projection and gallery	the first-floor projection 1.55 m over the colonnade (B3 / B4 on 4.33 + 6.25 m spans); the gallery behind W1 (sill +3.82), the family living behind W2, the living's upper side W3; the sky opening 3.15 wide between the edge beam and the kitchen wall	sections.typ 38–53 · structure.typ 73–74
Drains	the floor −0.015 at the three glass lines and the riwaq door, 1 % falls to −0.06 at the kitchen wall foot; two slot drains (along the kitchen wall foot y 9.6–12.8; at the riwaq door y 9.0–10.6), silt baskets, trap primers; the overflow gully at (8.6, 12.6) rim −0.03; the invert −0.55, 1 % east under the kitchen to the rain chamber (never foul)	character.yaml 153–162 · plumbing.typ 79, 328
Lighting	two hanging lanterns under the soffit edge between the columns (at y 9.75 and 16.125, 0.6 m drops); the warm room lights seen through the glass (the night view); nothing else designed	character.yaml 170–173, 202–205
Power and water points	the pump in the enclosure's divider (its circuit not drawn); the TV's power / data removed (10 Nov 2026); no sockets or lighting circuit for the courtyard designed yet; the malqaf chamber: IP44 sockets, a capped water point, the gully	courtyard_layout.typ 187 · air_system.typ § 2
Floor	outdoor anti-slip porcelain (R11) in a sand / honey tone, no pattern designed; level, no step;	DESIGN_LANGUAGE.md · character.yaml 153

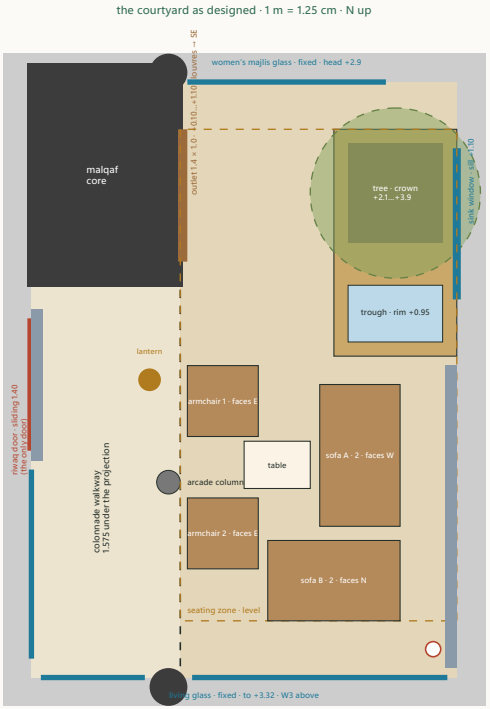
	trough's rim and under the crown	
Malqaf head	1.6 × 2.4 on the roof, x 4.3–5.9 / y 6.4–8.8, openings +11.2…+12.6 on all four faces (3 / 5 slots), the crown +13.2; the W and N faces the catchers; D1 at the shaft's head +7.75, D2 behind the outlet, spring-return, one switch	air_system.typ 27–28, 36, 203
Water + tree enclosure	one box on the kitchen wall x 7.55–8.85 × y 7.1–9.5 (walls 0.15): the planter 1.0 × 1.05 inside, 0.70 deep, rim +0.45 (north, in front of the sink window); the raised trough 1.0 × 0.6 inside, 0.35 deep, rim +0.95, water +0.85, a laminar lip at +1.05 (south); the pump chamber in the divider; both basins tanked, each to the rain system	character.yaml 145–152 · courtyard_layout.typ §3
The tree	**ASSUMED**: **Laurus nobilis** standard or clipped **Ficus nitida**, "a Taif nursery decides"; drawn stem 2.0, crown centre +3.0, radius 0.9 (bottom +2.1, top +3.9); drip irrigation; no root barrier noted	character.yaml 152

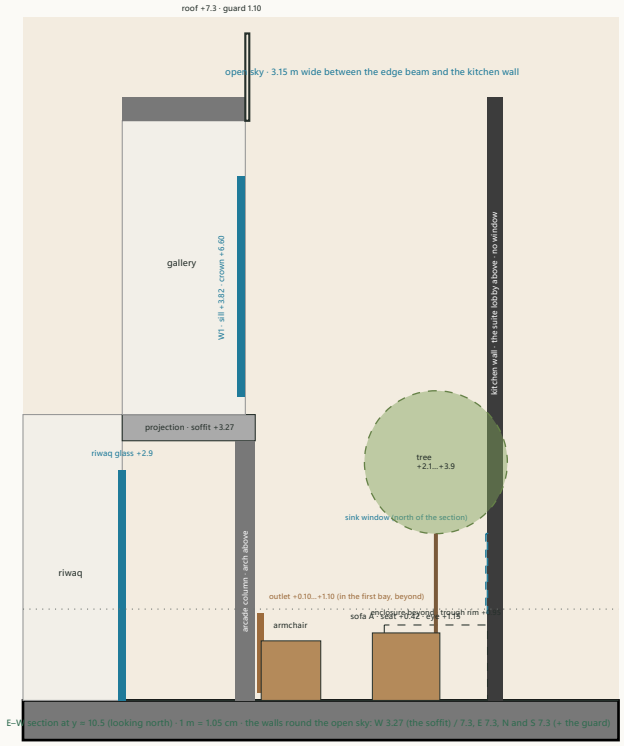
	the thresholds 15 mm at the glass	
Wind	**ASSUMED from the public station**: the warm-season evening wind W ≈ 50 %, NW ≈ 30 %, mean ≈ 21 km/h at 10 m; the house's own wake and the old house west of it estimated, the on-site station still to confirm	malqaf_L5.typ 23–30, 147
Heat and the seating hours	**ASSUMED**: the seating hours 16–23: pleasant and dust-free ≈ 1 800 h / year; > 32 °C in 14–36 % of the hours May–Sep; dust ≈ 60 days / year	malqaf_L5.typ 44–48
Sun	**ASSUMED**: Taif 21.3° N; solar noon ≈ 12:25 local; the sun's altitude at the summer solstice noon 92° − 21.3° ≈ 89° (overhead), at the winter solstice ≈ 45°; Phase 1 computes the shadows	geometry

Every opening onto the courtyard

Opening	What	Where
women's majlis glass (north, y 6.5)	x 6.0–8.1, floor to +2.9, fixed (no door)	presentation.typ 194 · params.yaml 47
riwaq glass door (west, x 4.25)	y 9.1–10.5, sliding 1.40: the only door into the courtyard	presentation.typ 173
riwaq glass (west, x 4.25)	y 10.7–12.7, floor to +2.9, fixed	params.yaml 50
riwaq's first panel (x 4.25, y 6.8–8.9)	the malqaf core's solid rendered face with a built-in bench (no glass since V10.0)	design_gf.typ 100 · PROJECT_V10 §4.63
living glass (south, y 13.0)	x 4.45–5.55 and 6.05–8.75, floor to the slab (+3.32), fixed (no door); W3 above it to the crown +6.60	params.yaml 48–49, 78
kitchen sink window (east, x 8.95)	y 7.3–8.9, sill +1.10, head +2.10; the enclosure in front of it	design_gf.typ 127 · character.yaml 144
malqaf outlet (west, the core's east face x 5.95)	y 7.1–8.5, +0.10…+1.10, timber louvres (free 60 %), the damper D2 behind	air_system.typ 30, 50, 203
malqaf door (the core's south face, y 8.767)	x 4.6–5.6, 1.0 × 2.4, carved timber, onto the covered strip (not the courtyard)	design_gf.typ 100 · air_system.typ 31
gallery windows above (west, x 5.8)	W1-2 y 8.95–10.55 and W1-3 y 11.117–12.717 (1.60 wide, sill +3.82, springing +5.80, crown +6.60); W1-1 (y 6.783–8.383) a blind arch	params.yaml 71–72 · ff_family.typ 124
family living windows above (north, y 6.5)	W2 × 2: x 6.0–7.3 and 7.55–8.85, floor +3.60 to the crown +6.60	params.yaml 76 · ff_family.typ 125
living's upper side (south, y 13.0)	W3: x 6.0–8.8 from +3.32 to the crown +6.60 over the living glass	params.yaml 78
kitchen wall above (east, x 8.95)	the suite lobby's wall, no window to the courtyard (the suite lobby F15 is blind to the court)	rooms_ff.csv, params.yaml windows

Phase 0 · the courtyard as designed: plan (clear faces) and the east–west section through the seating · the summary





Phase 0 · summary

- **The courtyard:** an L of 27.19 m² on axes, ≈ 24.88 m² clear, 4.50 wide between the riwaq glass and the kitchen wall, 6.30 long; the open sky 3.15 × 6.5 m between the projection's edge beam and the kitchen wall, 7.3 m deep (the roof) on three sides and 3.27 m (the soffit) on the colonnade side. The floor level, the porcelain finish, falls of 1 % to the kitchen wall foot.
- **Who opens onto it:** the women's majlis (fixed glass, north), the living (fixed glass to +3.32, south, W3 above), the riwaq (the sliding door 1.40 and fixed glass, west), the kitchen's sink window (east, +1.10…+2.10, behind the enclosure), the gallery's W1-2 and W1-3 (west, above the colonnade), the family living's W2 × 2 (north, above the majlis glass); the suite lobby above the kitchen wall is blind. **One door only** (the riwaq); the malqaf door opens onto the covered strip, not the courtyard.
- **Air:** the malqaf core fills the arcade's first bay; the outlet A low on its east face, 1.4 × 1.0, louvres to the south-east, ≈ 1.79 m³/s at 2.1 m/s in the mean summer-evening wind (W / NW); B would raise and turn it toward the trough. D1 / D2 spring-return on one switch.
- **Water and tree:** one box on the kitchen wall in front of the sink window; the tree north (planter 1.0 × 1.05 × 0.7, rim +0.45), the raised trough south (1.0 × 0.6 × 0.35, rim +0.95, water +0.85, a laminar lip +1.05), a pump in the divider, both basins to the rain system. The species **not decided** (laurel or clipped ficus), the canopy drawn Ø1.8 at +2.1…+3.9.
- **Seating:** six seats with backs (two 2-seat sofas, two armchairs, a table) in the level zone 2.925 × 5.2 (12.09 m² free), facing the outlet arch and the enclosure; no TV.
- **Services:** two slot drains and an overflow to the rain chamber; two lanterns under the soffit; the pump's circuit, sockets and a lighting circuit **not designed**; the chamber's maintenance door on the covered strip.
- **Assumptions to carry:** the wind (station data, the on-site station pending), the tree's size and species, the sun geometry (computed in Phase 1), every air figure (the MEP engineer).

Phase 0 checks passed (the facts read from the files: the courtyard polygon 27.19 m², the zone 12.09 m², the outlet 1.4 × 1, the enclosure 1.3 × 2.4)

Phase 1 · the 22 checks (and the owner's two notes) · pass / fail / unknown, with the numbers

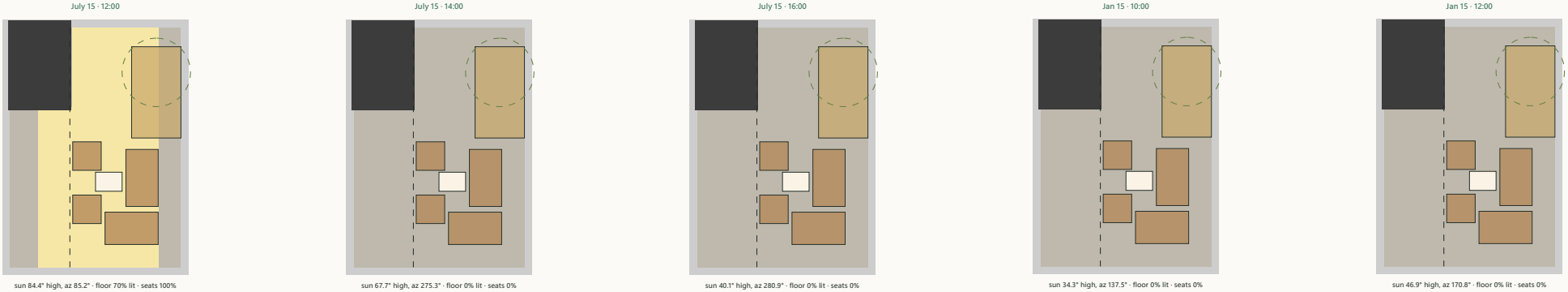
Sun: NOAA solar position for Taif (21.3° N, 40.4° E, UTC+3); a point is in sun when its ray to the sun passes the 3.15 × 6.5 sky opening at +7.3; the seats tested at +0.45 (the cushion), the eye at +1.15. Air: the free-jet estimate of `studies/malqaf_outlet_B.typ` (±30 %). **Amber = assumed.** A = the layout of 10 Nov 2026; B = the outlet study.

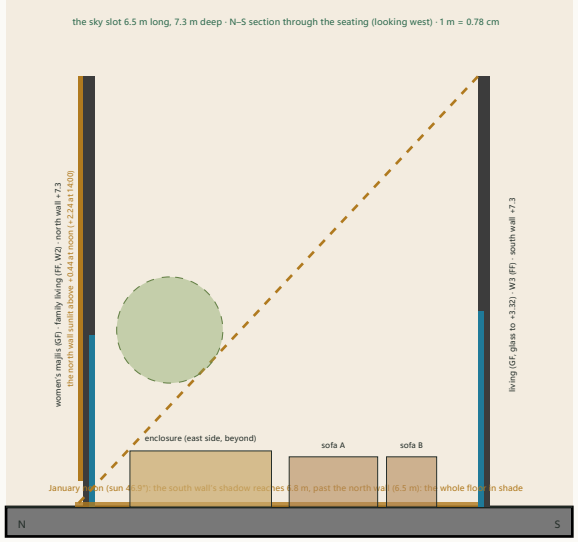
Check	The numbers	Verdict
Air		
1 · the breeze passes over the water and / or under the tree before the seats	A: the outlet +0.10…+1.10, louvres to the south-east: the jet runs over the floor to sofa A and the armchairs; the trough's rim is +0.95 and the crown's bottom +2.1: it crosses neither. B (studied): 59 % of the jet's depth over the water, its top under the crown.	A: FAIL · B: PASS
2 · the seats downstream at seated head height, no jet in anyone's face	A: sofa A faces the outlet 1.95 m away: the jet's top there is +1.44 with ±10° spread, the seated eye +1.15: **the jet is in the face**; armchair 1 at 0.45 m likewise. B: the sofa is 2.0 m off the jet's axis (half-width 1.2): the spill from the kitchen wall only.	A: FAIL · B: PASS
3 · the head faces the summer evening wind	the head's W and N faces catch the W (≈ 50 %) / NW (≈ 30 %) evening wind; the house's own roof wake and the old house to the west assumed until the on-site station.	PASS (wind assumed)
4 · hot air can rise out	the sky opening 3.15 × 6.5 = 20.5 m² over a 24.9 m² floor; the colonnade and the living's double height feed it; the stack ≈ 0.49 m³/s in calm.	PASS
Sun and shade		
5 · shadow on the seating at 16:00 and 18:00 in July, at 10:00 in January	July 16:00 (sun 40° high, WNW): the seats **100% shaded**; 18:00 (13°): 100 % shaded; July 14:00: 100% shaded; July noon (84°): 100% of the seats in sun. January 10:00 (34°, SE): 100 % shaded; January noon (47°): 100 % shaded.	summer PASS · winter FAIL (no sun)
6 · the tree's canopy shades in summer, lets winter sun in	the crown Ø1.8 at +2.1…+3.9 over the planter in the north-east corner: it shades 2.5 m² of the enclosure and the wall, none of the seats (they are 0.3–3 m south-west of it); in winter no sun reaches the floor anyway (5): the canopy is irrelevant to both.	FAIL (no effect)
Water		
7 · moving water, no mosquitoes; heard from the seats; safe for children	a laminar lip +1.05 into the trough (water +0.85): a 0.2 m fall, moving while the pump runs (it must run daily: no standing water). Heard: a laminar sheet is nearly silent (≈ 35–40 dB(A) at 1 m, assumed): not heard from the seats 2.5–4 m away. Safe: a raised trough (rim +0.95) 0.35 deep: a small child cannot fall in; the planter 0.7 deep is soil.	moving PASS · heard FAIL · safe PASS
8 · leaves in the water and their cleaning	the crown's edge (y 8.675) is 0.08 m from the trough (y 8.75): the leaves fall straight into the water; a laurel sheds a little all year, a ficus a lot (and fruit). The trough 1.0 × 0.6 is skimmed by hand; no skimmer or cover.	FAIL
9 · roots vs the water, the drains and the waterproofing	the planter 1.0 × 1.05 × 0.7 (≈ 0.7 m³) sits in a 0.75 recess on the tanked courtyard slab beside the trough and the main drain; a **Ficus nitida** (aggressive, invasive roots) would fill it in 3–5 years, crack it and block its drain: **unsuitable**; a laurel (fibrous roots) stays, but a standard tree in 0.7 m³ stays small (≈ 3–4 m) and needs irrigation every day in summer. No root barrier is drawn.	ficus FAIL · laurel PARTIAL
Use		
10 · every seat sees the tree or the water, not a wall or a service door	sofa A faces west: the core's rendered face and the outlet arch (a wall, 1.5 m away), the enclosure only over its right shoulder; sofa B faces north: the tree and the water ✓; armchair 1 faces east: the enclosure ✓; armchair 2 faces east: the kitchen's plain wall, the enclosure obliquely.	PARTIAL (2 of 4)
11 · conversation: the farthest seat-to-seat < 3.5 m, nobody's back to the group	farthest 2.23 m (sofa B ↔ armchair 1); every piece faces the table; sofa B's back to the living glass, sofa A's to the kitchen wall: the group is a square round the table.	PASS
12 · clearances: the walkway, the doors, the glass cleaning, behind the sofas	the colonnade walkway 1.575 clear (nothing west of the soffit edge); 0.6 at the living glass, 0.5 at the majlis glass, 0.6 behind sofa A at the kitchen wall; the riwaq door's 1.40 opening free; 0.3 only between sofa A and the enclosure (the trough's rim to clean).	PASS (0.3 at the trough: tight)
Comfort and calm		
16 · winter afternoons: sun on the seats, shelter from the cold wind	**no direct sun on any seat on any winter day** (5): the south wall (the living, 7.3 m) shades the whole floor from October to February; at January noon only the north wall's face above +0.55 is sunlit (the women's majlis glass), at 14:00 above +2.32 a seat with its back to the north glass would have its body in the sun for ≈ 2 h round noon. Shelter: the breeze shut by the switch; the well is still air ✓.	sun FAIL · shelter PASS
17 · night use: soft lighting, no glare into the rooms	two hanging lanterns under the soffit, the room lights through the glass; no lighting circuit, no sockets, no fittings in the enclosure or the water designed: unresolved.	UNKNOWN (not designed)
18 · acoustics: the water's sound, the fan's noise, echo	the lip is nearly silent (7); the malqaf's fan behind the core's 0.30 RC wall with its silencer (≈ 38 dB(A) in the chamber): inaudible ✓; the well: 7.3 m of render and glass on four sides over a porcelain floor, ≈ 20 m² of soft surfaces only (the sofas, the tree): a hard, reverberant room (RT ≈ 1.5–2 s): voices ring, the street's sounds are shut out.	fan PASS · water FAIL · echo PARTIAL
19 · scent and planting without mess	one tree only; a laurel gives scent (its leaves) and little mess; a ficus gives none and much mess; no other planting, no climbers on the 7.3 m walls, no pots.	laurel PARTIAL · ficus FAIL
Proportion and beauty		
20 · the proportions (width : height, the sky view), the rhythm, the view from each doorway	the sky slot 3.15 wide between walls 7.3 high: **1 : 2.3** (the courtyard floor 4.5 wide: 1 : 1.6); the traditional comfortable range 1 : 1 to 1 : 1.5. From a seat (eye +1.15) the sky is a patch 29° × 55° (the sky view factor ≈ 0.07 a good courtyard 0.2–0.4): **a deep well**. The rhythm: three arches of 2.17 on the west (the first filled by the core), W1 above them aligned, W2 × 2 north, W3 south: coherent ✓. From the riwaq door you face the kitchen's plain wall 4.5 m away with the enclosure in the corner; from the living's glass you face the majlis glass and the back of sofa B.	FAIL (the well) · rhythm PASS
Build and upkeep		
21 · drainage; power only where needed; the malqaf's maintenance door clear; cleaning the floor, the water and the leaves	the falls, two slot drains, the overflow and the rain chamber ✓; the malqaf door on the covered strip, 1.0 m clear ✓; the level porcelain floor hosed to the drains ✓; the water and the leaves: by hand, daily in autumn (8); power: nothing designed (17).	drains PASS · leaves FAIL · power UNKNOWN
22 · the evaporative cooler option	the space 0.8 × 0.7 × 1.6 in the chamber, its outlet against the louvres, a socket, the floor drain, the water point capped ✓; it would blow through the same louvres (A: into sofa A's face at 1.5–2 m).	PASS (A: the same face problem)
The owner's two notes		
23 · one door only: would a door from the living and / or the women's majlis make the courtyard far better used?	today the living's 3.8 m of glass (x 4.45–5.55, 6.05–8.75) and the majlis's 2.1 m are fixed: from the living's centre the seating is reached in 9.5 m via the riwaq door, 4.9 m through a door in the living's glass. **The living:** the family's main room, double height on this side, the glass to +3.32 under W3: a sliding door 1.40 at x 6.05–7.45 is simple (the frame only; the beam R2 above is untouched), the threshold 15 mm, and the courtyard becomes the living's outdoor room — the strongest single gain found. **The women's majlis:** a door would hand the courtyard to the	living door: FAIL today, a clear gain · majlis door: no

13 · privacy from the guest route and the majlis door	the men's guest route (foyer → majlis) has no view of the courtyard ✓; the **women's majlis** looks straight into the courtyard through 2.1 m of fixed glass (x 6.0–8.1, floor to +2.9): women guests see the family's seating 0.6–6 m away, and the family sees them; no curtain or screen is designed.	guests **PASS** · women's majlis **FAIL**
14 · the guest route never crosses the family seating	the courtyard has one door, from the riwaq (family side); the guests never reach the riwaq.	**PASS**
15 · the views into the courtyard from the living, the women's majlis, the kitchen sink, the gallery	the living (south glass to +3.32): the whole courtyard, sofa B's back in front; the women's majlis (north glass): the sofas' fronts and the core; the sink (window +1.1…+2.1): the trough's rim and the tree, as designed ✓; the gallery (W1-2 / W1-3, sill +3.82): down onto the seating and the enclosure ✓; the family living (W2): down onto the enclosure ✓; the suite lobby: blind.	**PASS** (the living sees a sofa's back)

	women's guests and remove the family's privacy (13): keep the fixed glass with a curtain.	
24 · the "deep well": the sky 3.15 × 6.5 inside 7.3 m walls	proportion 1 : 2.3 and the sky view ≈ 0.07 (20); no winter sun on the floor from October to February (5, 16); summer shade from 13:30 (good); the slot runs north–south, so the only sun that reaches the floor comes near noon. Within the open items the well stays: the walls are rooms (the projection, the family living, the kitchen and the suite lobby above); what can change is how it is lived with: the light colour and the lanterns, a taller tree reaching into the slot, a bright floor, the seating against the north wall for the winter noon, the living's door. Bolder ideas go to Phase 3's "ideas for the owner".	**FAIL** (inherent) · mitigations in Phase 3

Phase 1 · the sun: where the floor and the seats are lit (the sky opening's shadow) · the N–S section at winter noon · the summary





Phase 1 · summary (15 verdicts fail, 16 pass, 2 unknown)

- **The biggest finding is the well.** The sky slot is 3.15 m wide between 7.3 m walls (1 : 2.3; the sky view from a seat ≈ 7 %): **no direct sun reaches the floor or any seat from October to February**; in summer the floor is shaded from about 13:30 (good). The walls are rooms, so the well stays; what the open items can do is live with it: the north wall's face is in the winter noon sun above +0.5, so a seat with its back to the women's majlis glass gets the sun on the body for ≈ 2 h; the light render, the lanterns, a taller tree in the slot, the living's door.
- **The air works but hits the faces (A):** the low outlet aimed at the seats puts the jet's top at +1.44 on sofa A's sitters (eye +1.15) and misses the water and the tree. The raised, turned outlet (B) fixes both: the air over the trough and under the crown, the seats in the spill.
- **The water and the tree are in the wrong relation:** the crown hangs 0.08 m from the trough (leaves in the water), the laminar lip is inaudible, the planter (0.7 m^3) is too small for a ficus (roots, the drains) and small for a laurel; no skimmer, no root barrier.
- **Use:** the conversation square and the clearances are right (11, 12); two of the four seats look at a wall (10); the women's majlis's fixed glass gives no privacy from women guests (13); **the living has no door** onto its own courtyard (23): 9.5 m round by the riwaq against 4.9 m through the glass; a sliding door at x 6.05–7.45 is the clearest single gain; no door from the majlis.
- **Not designed yet:** the lighting and the power (17, 21); the acoustics are hard (18); the planting is one tree (19).
- **Phase 2** ranks these; **Phase 3** draws 2–3 layouts within the open items (the outlet B, the seating turned to the water and the north wall, the enclosure re-made with the tree away from the water, a skimmer, the living's door, the lighting, the floor pattern, the planting) and lists the bolder ideas (the well) for the owner.

Phase 1 checks computed (the sun at 10 moments; the seats' lit shares; the jet geometry; the paths).

Phase 2 · the mistakes, ranked · why each matters and how big it is · what the open items can fix

Ranked as the owner asked: the winter sun and the deep well first, with the air in the face; then by how much each spoils the haven. "Within the open items" means Phase 3 can fix it without touching the locked structure and rooms; the first two cannot be fixed that way, only softened, and their bold answers go to the owner in Phase 3.

№	The mistake	Why it matters	Its size	Fixable within the open items?
1	**No winter sun on any seat, October to February** (checks 5, 16)	The haven's best season in Taif is the cool half of the year (the pleasant seating hours peak Oct–Apr, `malqaf_L5.typ` 44–46); a courtyard without winter sun is a cold, grey room for five months.	100 % of the seats shaded at 10:00, 12:00, 14:00 and 15:00 on 15 Jan; the only sun on the north wall's face above +0.46 at noon	**not within the open items** (the walls are rooms): Phase 3 gives the mitigation (the north-wall seat) and the bold ideas for the owner
2	**The deep well** (checks 20, 24)	A sky slot 3.15 m wide between 7.3 m walls (1 : 2.3) and a sky view of ≈ 7 % from the seats: the courtyard reads as a shaft, not a garden; it also makes 1 inevitable and the echo (18) worse.	proportion 1 : 2.3 (the good range 1 : 1–1 : 1.5); sky view factor 0.07 (0.2–0.4 in a good courtyard); the sky a 29° × 55° patch	**not within the open items**: mitigations only (the colour, the lanterns, a taller tree, the living's door); the bold ideas in Phase 3
3	**The breeze blows into the faces and misses the water and the tree** (checks 1, 2)	The malqaf is the courtyard's reason: a 2 m/s jet at +0.1…+1.4 on the people facing it 2 m away is a draught, not a breeze; and the water and the tree, the two things that cool and scent the air, are beside the jet, not in it.	sofa A's eye +1.15 inside the jet (top +1.44 at 1.95 m); 0 % of the jet over the water or under the crown	**within the open items**: the outlet raised and turned (B), the seats in the spill, 0.6 m/s instead of 0.8–1.0 but on the whole body
4	**The living has no door onto its own courtyard** (check 23)	The family's main room looks at the haven through 3.8 m of fixed glass and reaches it round two corners; the courtyard is used less than it could be and the living never spills out.	9.5 m via the riwaq door against 4.9 m through the glass; 1 door today, 2 with it	**within the open items**: a sliding door 1.40 in the living's glass (frame only)
5	**Leaves in the water; the tree's roots beside the water, the drain and the waterproofing** (checks 8, 9)	A trough skimmed by hand every day in autumn is a chore the family will drop; a ficus in a 0.7 m³ built-in planter breaks it within years.	the crown's edge 0.08 m from the trough; the planter 0.7 m³ against 2–3 m³ for a tree; no barrier, no skimmer	**within the open items**, and **settled by the owner's olive in a fixed pot** (below), away from the water; the enclosure re-made
6	**The women's majlis looks straight into the family's seating** (check 13)	Whenever women guests sit in the majlis, the family's courtyard is on view through 2.1 m of clear glass 0.6–6 m away, and the guests are on view from the courtyard: the haven stops being private on every reception day.	glass x 6.0–8.1, floor to +2.9, fixed, no screen	**within the open items**: a curtain or a timber lattice on the majlis side (the design language's mashrabiya), or obscured glass below +1.6
7	**The water is silent** (checks 7, 18)	The sound of water is half of what a courtyard's water gives; a laminar lip over a 0.2 m fall is inaudible from 3 m.	≈ 35–40 dB(A) at 1 m, assumed; the seats 2.5–4 m away	**within the open items**: a higher fall (0.4–0.5 m) into a deeper basin or a small spout; the pump sized for it
8	**Two of the four seats look at a wall** (check 10)	Sofa A faces the core's rendered face 1.5 m away; armchair 2 faces the kitchen's plain wall: the two things worth looking at (the tree and the water) are behind or beside them.	2 of 4 seats; sofa A's view 1.5 m of wall	**within the open items**: the seating turned to the water and the tree (B's arrangement or the north-wall seat)
9	**The hard well: echo** (check 18)	Four hard faces 7.3 m high over porcelain: voices ring; the water's sound (7) would help, and so do the sofas, the tree and a soft wall.	≈ 20 m² of soft surfaces in a 24.9 m² room 7.3 m deep; RT ≈ 1.5–2 s (assumed)	**within the open items**: planting on a wall, fabric, the water's sound; the colour
10	**Lighting and power not designed** (checks 17, 21)	Two lanterns and the room lights are not a night courtyard; the pump, the sockets, the lighting circuit and its switch exist nowhere.	0 fittings designed, 0 circuits	**within the open items**: a lighting plan (soft, low, no glare into the rooms), a socket pair, the pump's circuit; the MEP engineer
11	**One tree, nothing else** (check 19)	A single tree beside the water is a token; scent, softness and the sense of a garden need more than one plant, without mess.	1 tree; 0 m² of other planting	**within the open items**: pots and a climber on one wall (Phase 3)
12	**The trough's rim 0.3 m from sofa A** (check 12)	Too tight to clean the rim or to walk round the sofa.	0.3 m	**within the open items**: the seating moved (3, 8)

Phase 2 · the olive in a large, free-standing, fixed pot (the owner's choice) · the pot, its best position, the weight, the water · the four checks re-run

The owner (10 Nov 2026): the tree is an **olive** in a large, free-standing, **fixed** pot (not in the ground, not a built-in planter): the roots contained; its pot, its one best position (summer shade on the seats, away from the water), its weight, its drainage and watering. Taif: 1 700 m, winter nights 2–5 °C with rare frost, summer days 32–36 °C, dry air, hard water (**station data; the nursery confirms the hardiness on site**).

Species	Why / why not in a pot in Taif	Pot	Weight	Verdict
Olive · *Olea europaea* (a fruitless or a table cultivar; a 2–3 m standard)	the right tree for a pot in Taif: hardy to −8 °C (the winters), drought-tolerant, evergreen with small silver leaves (little litter, light under it), slow and happy in a pot for decades, the classic Hijaz / Mediterranean courtyard tree; no scent (a laurel or a lemon beside it for that); a clipped head 2–2.5 m across at +1.8…+3.9	Ø 100 × 80 cm (≈ 0.55 m^3 of soil)	≈ 1.2 t wet (soil 0.9 + pot 0.15 + tree 0.1)	**the owner's tree**
Bay laurel · *Laurus nobilis* (a clipped standard or cone)	hardy to −5 °C, evergreen, dense, the leaves' scent when brushed, little litter; slower, wants a little shade from the July sun on its pot; the second tree or the alternative	Ø 80 × 70 cm (≈ 0.3 m^3)	≈ 0.7 t	**a second pot (scent)**
Lemon · *Citrus limon* (a dwarf or a Meyer on a dwarfing rootstock)	the scent of the blossom and the fruit are the courtyard's pleasure, but it is frost-tender (Taif can touch 0 °C in January: a fleece on the coldest nights), drops leaves and fruit, wants feeding and steady water; keep it as a **third, smaller pot** in the sunniest spot (the north wall's foot at noon), not as the main tree	Ø 60 × 55 cm (≈ 0.15 m^3)	≈ 0.35 t	**optional, winter cover**

The olive's pot, its fixed position and its services

- **The pot:** Ø 100 cm, 80 cm deep (≈ 0.55 m^3 of soil), glazed terracotta or GRC in the sand / honey tone (the render-and-paint language), **fixed** on pot feet 5 cm high so the base dries; 4–6 drain holes over a **saucer with an overflow spout** led by a Ø 32 pipe in the floor build-up to the nearest slot drain; the soil a free-draining olive mix (loam, sand, 30 % perlite) over 10 cm of gravel.
- **Its one best position: inside the seating group at its south-west corner, against the colonnade's soffit edge** (the pot's centre ≈ x 6.5, y 11.8: east of the soffit edge x 5.925 so the walkway stays clear, north of the living's cleaning strip y 12.3–12.9, ≥ 2 m from the water): near the summer noon the sun stands almost overhead (84° in July), so a tree shades only what is **under its head**: from this corner the 2 m head (+1.8…+3.9) covers the table and the near ends of both sofas, ≈ 3 m^2 of the seats, at the only hours the walls do not shade them (11:00–13:30); its leaves fall on the floor by the colonnade, never in the trough (3 m away); it is the first thing seen from the riwaq door and from the living's glass, and it reads with the arcade behind it. In winter it stands in the shade like everything else (1). Phase 3 draws the seats round it.
- **Water:** a drip ring on a timer from the enclosure's supply (the pump chamber's water point, a buried Ø 16 line to the pot): ≈ 15–25 L twice a week in summer, every 10 days in winter (**assumed; the nursery sets it**); the saucer never standing; a yearly feed; the head clipped once a year.
- **Weight (a note for the engineer):** ≈ 1.2 t wet on a Ø 1.0 m base ≈ 1.5 t/m^2; the courtyard slab is ground-bearing (the house's ground slab, lowered 0.05 for the outdoor build-up, `studies/plumbing.typ`): fine; if the engineer makes the courtyard a suspended slab over a drainage void, the pot's spot is a point load to note.
- **Fixed:** no castors; the pot's circle marked in the floor pattern (Phase 3); the saucer's pipe permanent (to the riwaq-door slot drain 1.6 m west or the kitchen-wall drain 2.2 m east, Phase 3 chooses); if the pot must ever move, a pallet trolley and two people.
- **Size:** buy it at 2.0–2.5 m with a 1.5 m head (a 15–20-year tree from a Taif or Hijaz nursery); it reaches 3 m in a few years and stays there with the clip; the head starts at +1.8 so the seated people sit under it, not in it; the pot's top at +0.85, no climbing edge for a child.

The four checks re-run with the pot

Check	With the pot	Verdict
6 · the canopy shades in summer, lets winter sun in	the olive fixed at the seating group's south-west corner, 2.5–3 m high with a 2 m head: at July noon (the sun 84° high) it shades ≈ 3 m^2 of the seats under its head, the table and the sofas' near ends, in the only sunlit hours (11:00–13:30; the walls shade everything after); in winter the floor gets no sun whatever the tree does (1), and the olive's thin evergreen head takes nothing away.	summer **PASS** · winter n/a
8 · leaves in the water	the pot stands ≈ 3 m from the water: the olive's small leaves fall on the floor by the colonnade and are swept; nothing hangs over the trough.	**PASS**
9 · roots vs the water, the drains, the waterproofing	the roots are inside the pot; the pot stands on the finished floor (no recess, the membrane untouched); its drain holes over a tray or a saucer piped to the nearest slot drain (the kitchen-wall drain, y 9.6–12.8); **the weight:** ≈ 1.2 t on a Ø 1.0 m base (≈ 1.5 t/m^2) at one fixed spot: the ground-bearing courtyard slab takes it; the engineer notes the spot (the slab is lowered 0.05 there for the build-up, `plumbing.typ`) and, if the courtyard ever becomes a suspended slab, designs for the point load.	**PASS** (engineer's note)
19 · scent and planting without mess	the olive gives no scent and little mess; the laurel gives scent when brushed; the lemon gives the scent and some mess; a jasmine on a wire on one wall (the kitchen wall's plain part) gives the evening scent with little litter. The olive fixed, a laurel and a lemon as smaller pots where they fit (Phase 3), a jasmine on a wire: a garden, not a token.	**PASS** with the pots

What the pot changes in the design

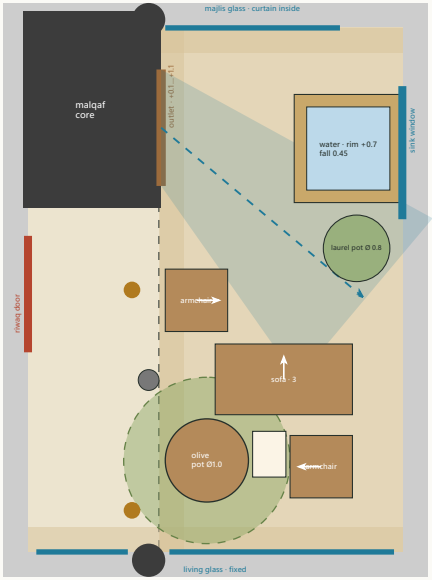
- The built-in planter in the enclosure goes: the enclosure becomes **the water only** (Phase 3 re-draws it: a trough or a basin with an audible fall, a skimmer, the pump), 0.8–1.0 m shorter along the kitchen wall; the sink's view becomes the water and, beside it, the olive's pot.
- The floor pattern marks the pot's circle and the furniture line; the pot's drain pipe and the drip line are small items for the plumbing study; a socket pair and the lighting (Phase 3) near it.
- Mistake 5 is settled; mistakes 7, 8 and 11 are opened for Phase 3's layouts.

Phase 2 complete: 12 mistakes ranked; the pot re-checks 6, 8, 9 and 19 → pass. Phase 3 next: the layouts and the ideas for the owner.

Phase 3 · three layouts within the open items · the owner's answers built in

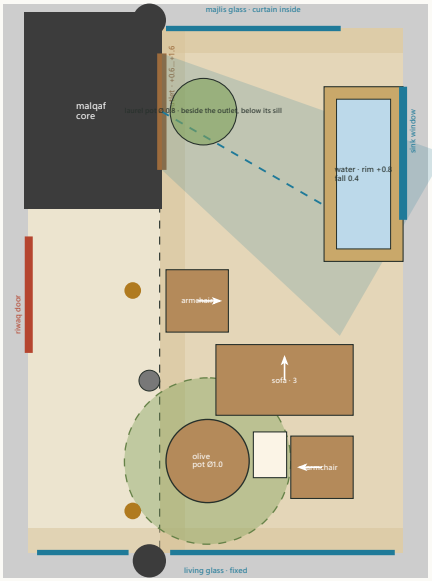
In every option: the living's glass fixed and the riwaq's sliding door the only door; a curtain inside the women's majlis; the water a calm, audible fall with a skimmer; the olive in its fixed Ø 1.0 pot at the seating group's south-west corner (x 6.0–7.0, y 11.3–12.3), the built-in planter gone; every seat with a back; the walkway, 0.6 at the living glass, 0.5 at the majlis glass (H3: a movable sofa stands there), 0.6 at the kitchen wall. The jet's axis dashed, its spread shaded; the lanterns amber; the darker floor bands on the glass lines and the colonnade line.

H1 · fixes in place



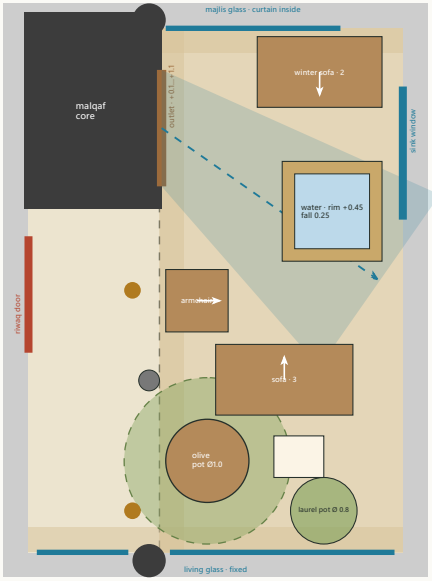
5 seats with backs · 1 m = 1.1 cm

H2 · the breeze over the water (outlet B)



5 seats with backs · 1 m = 1.1 cm

H3 · the water in the middle, a winter seat on the north wall



6 seats with backs · 1 m = 1.1 cm

H1 · fixes in place the outlet A kept (its louvres set low, along the floor); the water re-made where it is; the olive; the seats turned to the water

- **Outlet:** A as built: low; the louvres fixed 40° south-east and 10° down so the jet runs along the floor to the kitchen wall, not at the faces
- **Water:** a basin 1.3 × 1.3 on the kitchen wall under the sink window: a wall spout at +1.00 falls 0.45 into water at +0.55, rim +0.70; a skimmer basket at the overflow; the pump in a dry chamber under the spout
- **Seats (5):** sofa · 3 facing N, armchair facing E, armchair facing W the table 0.4 × 0.55 under the olive's head.
- **Light:** the two lanterns (soffit) + a submerged warm light in the basin + two low wall lights on the kitchen wall's plain part (a 2700 K strip behind the jasmine wire); one switch by the riwaq door, one dimmer
- **Floor:** plain sand porcelain, a darker 0.3 m border on the three glass lines and the colonnade line; the olive's circle and the furniture line in the darker tone
- **Planting:** the olive; a laurel pot by the basin (scent); a jasmine on a wire on the kitchen wall south of the basin
- **Changes outside the courtyard:** `studies/air_system.typ` (the louvres' fixed angle only), the courtyard study and `3d/character.yaml` (the enclosure re-made, the furniture, the pot), the plumbing (the basin's pump chamber, the skimmer, the pot's drain and drip line), the lighting circuit and two sockets (MEP), the 3D, sheet 02

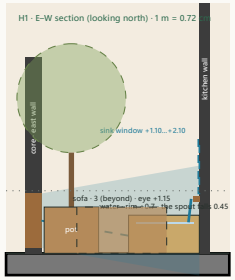
H2 · the breeze over the water (outlet B) the outlet raised and turned at the water; the water a long trough along the kitchen wall in the jet's path; the seats in the spill facing the water

- **Outlet:** B: raised, centred in the bay, the louvres fixed 30° south-east with splayed reveals: over the trough's rim and under the olive's head
- **Water:** a trough 2.1 × 0.95 along the kitchen wall under and south of the sink window: a wall spout at +1.05 at its north end falls 0.40 into water at +0.65, rim +0.80 (the sill +1.10 above it); the skimmer at the south end where the flow ends; the pump in a dry chamber
- **Seats (5):** sofa · 3 facing N, armchair facing E, armchair facing W the table 0.4 × 0.55 under the olive's head.
- **Light:** as H1, plus a light under the trough's spout; the olive up-lit from the floor
- **Floor:** a porcelain "carpet": a 0.6 m border in the honey tone round the seating zone, the field sand; the olive's circle in the border tone
- **Planting:** the olive; a laurel pot in the north-west corner (its head below the raised outlet's sill +0.6: it does not block the jet); a jasmine on the kitchen wall's south part
- **Changes outside the courtyard:** `studies/air_system.typ` (`air-outlet` → y 6.9–8.3, +0.60…+1.60, fixed louvres), `3d/params.yaml` (the outlet window), the courtyard study and `character.yaml` (the trough, the furniture, the pot), the plumbing, the lighting and sockets, the 3D, the DXF, sheet 02

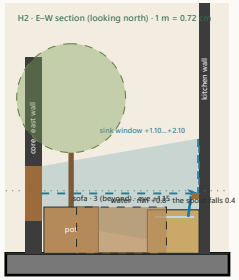
H3 · the water in the middle, a winter seat on the north wall the water a low basin in the courtyard's north-east quarter; a 2-seat sofa with its back to the women's majlis glass for the winter noon; the outlet A re-aimed at the basin

- **Outlet:** A as built, the louvres fixed 35° south-east at the basin (the jet skims the water: cooled a little) and clear of the winter sofa
- **Water:** a low square basin 1.2 × 1.2, 0.25 m off the kitchen wall, rim +0.45, water +0.30 (a mesh 5 cm under the surface for the children), a central calm bubbler 0.25 high: audible from every seat; the skimmer basket in a corner; the pump below the basin
- **Seats (6):** winter sofa · 2 facing S, sofa · 3 facing N, armchair facing E the table 0.6 × 0.5 under the olive's head.
- **Light:** as H1, with the bubbler lit from below and a low light at the winter sofa; the lanterns dimmed; nothing shining into the majlis or the living
- **Floor:** the field sand; a honey-tone square frames the basin (0.4 m) and a band marks the winter sofa's place at the north glass; the olive's circle
- **Planting:** the olive; a laurel pot in the south-east corner by the living glass (moved for cleaning); a jasmine on the kitchen wall's south part; the basin's margin a single potted papyrus or none
- **Changes outside the courtyard:** `studies/air_system.typ` (the louvres' fixed angle), the courtyard study and `character.yaml` (the basin, the winter sofa, the furniture, the pot; the north cleaning strip given up for a movable sofa), the plumbing (the basin, the pump, the skimmer, the pot), the lighting and sockets, the 3D, sheet 02; the majlis curtain (the interiors)

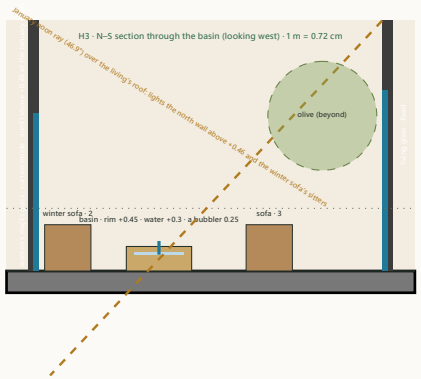
Phase 3 · the sections · the ideas for the owner (winter sun and more sky: bold, touching locked items)



E–W through the seating: the low jet runs under the seated eye to the wall; the basin's fall 0.45; the olive's head over the table



E–W through the outlet's axis: the jet +0.6…+1.6 over the trough's rim +0.80 and under the olive's head +1.8; the spill down the kitchen wall onto the sofa



N–S through the basin: the winter sofa's back on the north glass (its body in the January noon sun above +0.46), the basin 0.8 m in front, the main sofa opposite; the olive's head at the right

Ideas for the owner · winter sun and more sky (touching locked items: listed, not drawn)

Idea	What	Knock-on effects (the locked items)	Verdict
I1 · a sun reflector on the living's roof edge	a polished aluminium "sun shelf" ≈ 3.0 × 0.8 m on the south roof edge (y 13, +7.4), tilted ≈ 23°: it throws the January noon sun (46.9°) down onto the courtyard's floor and the seats as a soft reflected patch (≈ 25–35 % of direct sun, **assumed**); summer: tilted flat or removed	none of the locked items; the roof's guard and parapet detail; glare checked from the gallery; cleaned yearly	cheap; the only idea that puts winter light on the seats without opening a roof; it does not widen the sky
I2 · open the living's double-height bay to the sky (a second court)	the roof over the living's double-height half (x 4.25–8.95, y 13.0–16.1, 14.6 m²) removed: the courtyard becomes 4.7 × 9.6 m on axes with the south wall at y 16.1; the January noon shadow (0.936 × 7.3 = 6.8 m) then ends at y 9.3: **every seat south of y 9.3 is in the winter noon sun**, the sky slot 3.15 × 9.6	LOCKED: the living loses its double height and its clerestories (R-24's "one living room, two heights" since V3), W3, the roof edge R2 and the slab bay over the void (structure), the TV wall's upper part, the suite court above? (no: the suite court is over the men's majlis); the living's glass becomes an outer wall; drainage of a second open floor	the biggest gain for the winter and the sky, the biggest loss for the house; a different house
I3 · drop the south wall's top to +3.6 only over the courtyard's width (a roof terrace over the living's void)	the living's double-height roof stays over the living, but its north edge is pulled back 2.0 m (y 13 → 15): the strip y 13–15 over the void becomes an open terrace of the suite court's level… no: the void has no first floor; the strip would be a roofless slot inside the living	LOCKED: the living's void and roof; a slot open to the sky inside a room is not a room	not workable: struck
I4 · lower the gallery projection (the west side) to the first floor	the gallery's roof over the projection removed, the gallery open to the sky as a first-floor loggia: the west wall of the slot drops from +7.3 to +3.6 (the guard); the afternoon sun from the south-west (January 15:00, 33°) still falls on the kitchen wall, not the floor (the shadow of +3.6 reaches 5.5 m east); the sky slot's west side opens: sky view ≈ 12 % instead of 7 %	LOCKED: the gallery (the family's corridor) loses its roof and its W1 windows: it becomes an open walkway in a Taif winter and dust; the roof slab over the projection (R1 / R3 / B3 upstand), the bedrooms' doors onto an open gallery	more sky, no winter sun; the gallery's loss is severe: not recommended
I5 · the family living above the women's majlis pulled back	the north wall (the family living's W2 wall, +7.3) could not help: the sun never comes from the north; only the slot's width matters, and it is fixed by the projection (west) and the kitchen wall (east)	—	no gain: struck
I6 · the winter haven elsewhere; the courtyard the summer haven	accept the well: the courtyard is the breeze, water and shade room for March–October; the **suite court** (first floor, open to the sky, 25 m², the parapet frame on its south) or the **roof terrace inside the attic** is the family's winter sun room (the suite court faces south over the men's majlis roof; the January noon sun stands 47° above it: fully sunlit), with its own simple seating	none of the locked items (the suite court's furniture and the attic terrace already exist in the plan); the family's habit, two places instead of one	honest and free; with I1 for the courtyard's winter light

What the three options do with the three big mistakes

- **The winter sun (1):** none of the layouts brings direct sun to the floor; **H3 alone** seats two people in it (the winter sofa with its back to the majlis glass: the body in the sun ≈ 11:30–13:30 from November to February, the curtain behind them). With I1 (the reflector) H1 and H2 get a reflected patch too.
- **The deep well (2):** all three lighten it the same way: the light sand render, the olive's head in the slot, the darker floor bands that pull the eye down, the lanterns at night, the water's sound; H3's basin in the middle makes the floor the room's centre instead of the walls.
- **The air in the faces (3):** H1 keeps the low outlet and sends the jet along the floor (the breeze at the legs, ≈ 0.5 m/s at the sofa, **assumed**); H2 raises and turns it over the water (≈ 0.6 m/s on the whole body, cooler); H3 skims the basin with the low jet (≈ 0.5 m/s, a little cooler) and keeps it off the winter sofa.
- **Seats:** H1 5 · H2 5 · H3 6 (the owner asked for 5–6, every one with a back).
- **The water's sound:** H1 a 0.45 fall, H2 0.40, H3 a bubbler: all audible from the seats (a fall ≥ 0.3 m into open water ≈ 45–50 dB(A) at 2 m, **assumed**).
- **Phase 4** scores all three against the 22 checks, the clear area, the seats, the relative cost and the files, and recommends.

Phase 3 complete: three layouts drawn (plans and sections), six ideas for the owner listed (two struck).

The geometry behind it: the January noon sun stands 46.9° high in the south; a wall H high throws a shadow 0.936 H long northwards. The seats are 1.65–2.5 m north of the living's wall (7.3 m): only a wall ≤ 2.3 m high there would let the noon sun on them — that is the living's roof. The slot's width (3.15 m) is set by the projection and the kitchen wall. So within the house as decided, the winter sun on the courtyard's floor is impossible; the reflector (I1) borrows it, the suite court (I6) has it.